

# WMS Build Roadmap

as of 2026-09-14

Warehouse Management System — build pipeline snapshot

Master Data → Yard & Gate → Inbound → Putaway → Inventory → Outbound → Picking → Dispatch → Returns → Analytics

■ Built ■ Started ■ Not started



## What's landed since the last refresh (2026-09-13)

A quiet couple of sessions compared to the ones before them — feature close-outs and one maintenance pass, no new modules.

### Inventory: Excel export shipped — module marked complete

**Line Items** and **SKU Summary** both gained an Export to Excel button (same convention the Ledger tab already had), closing the module's one remaining open item. The daily-volume dashboard's **outward** half was discussed and deliberately NOT built — genuinely blocked on Picking/Dispatch being designed first, not just deferred; that item now lives under **Analytics**, not here.

### Stillage: row-axis placement + Insights coverage

Finishes the same day's Stillage Putaway build — a real row-axis convention (mirroring Rack Rank/Ground's own **numberOneNearDock**-driven treatment) now factors into which Stillage position gets suggested first. The Insights "Storage Utilization by ABC Class" report was extended to cover Stillage lanes alongside Rack — **Ground/Floor stays excluded** (never asked for, not decided against).

### Weekly health-check pass

Backend/frontend lint cleanup, a real **AuthUser** type replacing roughly 330 hand-typed user: any parameters across the backend, and dependency/security patches (a multer CVE fix). Pure hygiene — no feature/behavior change, no new schema.

### Next up, per the stated build order

**Outbound → Picking → Dispatch** are the next real modules — everything built so far (Master Data → Yard & Gate → Inbound → Putaway → Inventory → Analytics-so-far) has been building toward them. No design conversation has started yet.

## Module-by-module status

| MODULE      | STATUS      | SUMMARY                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Master Data | Built       | Warehouses (network-node master: Factory/DC/CNF/Cross-dock), SKUs, Customers, Locations (with a full 3D Plan View), Users — all with role-scoped access and standard import/export.                                                                                                                                                                                                                                                                                      |
| Yard & Gate | Built       | Gate In/Out, dock assignment, document checks, real compass-direction dock zones + a live-rotating 3D compass overlay. One competitor-research pass done (self-check-in + physical inspection flagged, not built).                                                                                                                                                                                                                                                       |
| Inbound     | Built       | Order maker + Excel bulk import + ERP push, vehicle-matched orders, scan-based receiving, Complete Inward Process/Dock Out, per-dock default staging locations.                                                                                                                                                                                                                                                                                                          |
| Putaway     | Built       | Real bin-suggestion logic for every storage type (Rack/SPR, Drive-in, Ground/Floor, Stillage) — ABC/FMS-aware, reservation-aware, dock-relative, and (Rack + Stillage) row/rank-aware. Pallet consolidation, Pick Face for SPR, operator-assignment fairness, Aisle/Level Rank visuals. <b>Open:</b> cancel path, queue ordering, Pick Face for Drive-in/Ground/Stillage, Stillage's own "closed column" lifecycle tracking (blocked on Picking writing real movements). |
| Inventory   | Built       | On-hand Line Items + SKU Summary views, a transaction-level ledger, and Excel export on all three views (with From/To Location) — all live-derived, no stored counters. The Daily Volume dashboard's outward half is a separate, still-open <b>Analytics</b> item, deliberately deferred until Picking/Dispatch are designed.                                                                                                                                            |
| Outbound    | Not started | Schema exists; order maker with capacity checks (weight + volume) and pick-list generation not yet designed. Next module in the stated build order.                                                                                                                                                                                                                                                                                                                      |
| Picking     | Not started | Nothing built yet. <b>Task Interleaving is flagged HIGH PRIORITY</b> for the moment this starts — a deliberate revisit of an earlier "let Analytics compare, decide later" call.                                                                                                                                                                                                                                                                                         |
| Dispatch    | Not started | Nothing built yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Returns     | Not started | Nothing built yet.                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Analytics   | Started     | Operator productivity (marrying/putaway time per operator, Pick Face trip time, abandoned-claim flags) plus Daily Inward Volume charts — this app's first real charting page. Distinct from the earlier one-off Insights page.                                                                                                                                                                                                                                           |

## Immediate candidates for the next session

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Live options on the table — not a forced order.

1. **Outbound / Picking / Dispatch — the actual next module in the build order.** Everything built so far (Master Data → Yard & Gate → Inbound → Putaway → Inventory → Analytics-so-far) has been building toward these three. No design conversation has happened yet — needs its own align-before-coding pass, same as every prior module.
2. **The reslotting/consolidation suggestion engine.** Genuinely unblocked — ABC classification, dock-relative placement, and FMS are all in place. Schema has sat ready; the actual detect-a-misplaced-SKU/suggest-a-target-bin algorithm and its daily job still need designing.
3. **Daily Outward Volume (Analytics)** — the mirror of the already-built Daily Inward Volume chart. Deliberately deferred, not forgotten: neither Picking nor Dispatch exists yet to say which movement(s) (PICK, DISPATCH) genuinely represent a warehouse-exit event — no new schema needed, just the design conversation that decides the semantics.
4. ~~Inventory: Excel export + module close-out~~ — **BUILT** (2026-09-14): Line Items/SKU Summary both gained export, matching the Ledger tab's own. Inventory's own scope is now fully built.
5. ~~Stillage's own Putaway logic~~ — **BUILT** (2026-09-13): a physical redesign (one row per real stack position), a new per-position bin-suggestion strategy, then row-axis placement and Insights Storage Utilization coverage (2026-09-13/14).
6. ~~ASRS as a storage type~~ — **REMOVED** (2026-09-13, the client's own call): a real ASRS installation runs its own dedicated WCS/WES software.
7. **Pick Face for Drive-in / Ground / Stillage** — SPR-only today, deliberately (finish one storage type properly before generalizing).
8. **Dock-out → Gate-out signal** — an active notification once paperwork time after dock-out has genuinely elapsed. Needs a real design conversation first.
9. **Dock + Staging + Yard visualizer** — paused mid-conversation; needs a spatial-data pass first (Dock Doors/Yard Slots have no position data today).
10. **A cancel/void path for a mistaken Gate In** — flagged, not designed.

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Also this session: a weekly hygiene pass (lint cleanup, a real **AuthUser** type replacing ~330 user: any parameters, dependency/security patches) — maintenance, not a roadmap item, so it isn't numbered above.

## Deferred, lower priority — per explicit client calls, don't build unprompted

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### From the top-WMS gap check (vs. Blue Yonder / Manhattan / industry norms)

- **Task Interleaving** — the one exception on this list: raise it the moment Picking starts, not just "someday."
- Cycle counting / physical inventory reconciliation — no workflow exists at all yet.
- 3PL / client billing — a real gap against the multi-company SaaS goal; no rate cards or invoicing today.
- Vendor/Supplier master data — receiving is tied to a Vehicle, not to a vendor.
- Kitting/Assembly logic — schema exists, zero logic built.
- Quality Control / Inspection workflow — no formal hold→inspect→release/reject module.

### Putaway follow-ons

- A cancel/exception path for a task that can't be completed.
- Correcting an already-completed mis-putaway.
- Real queue-ordering / aging-based task prioritization.
- Pick Face for Drive-in / Ground / Stillage (SPR-only today, deliberately).
- Stillage's "closed column" lifecycle tracking — blocked on Picking ever writing real `MovementType.PICK` movements, not decided against.
- Pallet reuse (needs Picking to ever deplete a load).
- Ground/Floor still excluded from the Insights Storage Utilization report (Rack + Stillage covered) — never asked for.

### Everything else explicitly parked

- Self-service driver check-in, Yard Plan View (no spatial data yet).
- Closing a short receipt (no discrepancy close-out path today).
- Batch/Lot + expiry capture — deferred to when Inventory's own design revisits it.
- Real GS1/unique-barcode parsing, camera-based scanning.
- Real SMS/Email/WhatsApp/Voice providers — every channel is a logging stub today.
- Dock Scheduling algorithm — only its eventual output has anywhere to land.
- Blacklist enforcement at Gate In, a general analytics dashboard, WMS/TMS integration — explicitly parked.
- SUPER\_ADMIN account creation, a configurable permission matrix, a company-admin invite-link flow.
- Inventory Control Policy (min/max, reorder point, FIFO/FEFO/LIFO), Opening Balance load.
- Dispatch-proximity distance calculation, real SAP/ERP integration, cloud/production deployment.